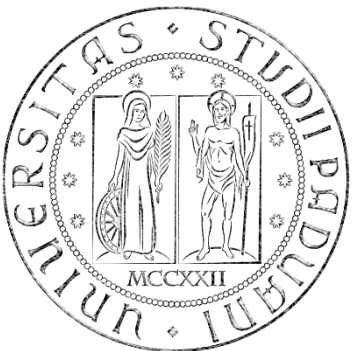


Biomedical Wearable Technologies
for Healthcare and Wellbeing

Fundamentals of computer networks and HTTP

A.Y. 2025-2026
Giacomo Cappon



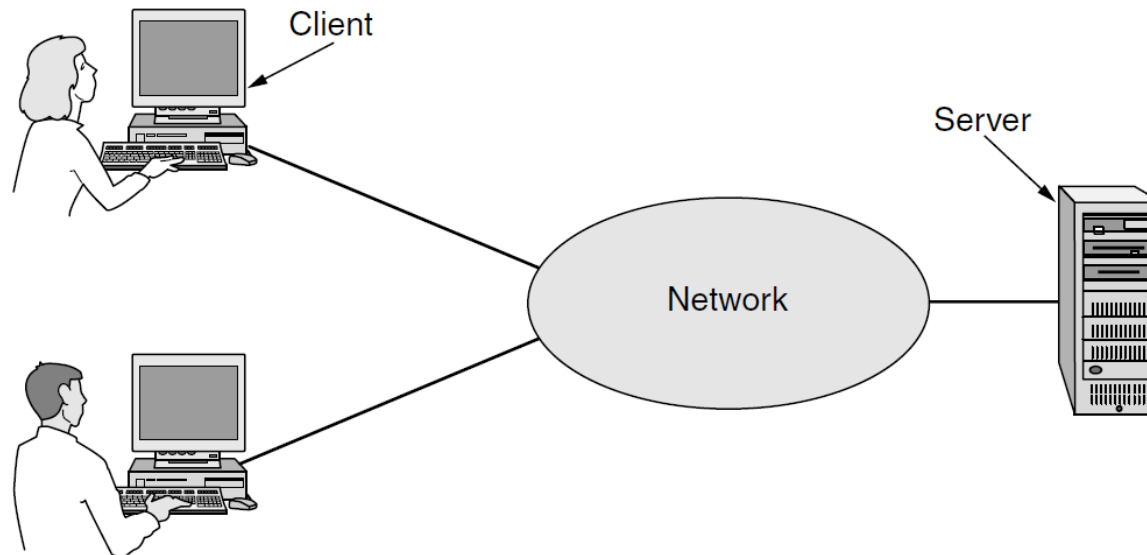
Computer networks

- Data collected by wearable IoT sensors are shared among different devices running over a computer network, typically the Internet.
- **Computer network:** a collection of autonomous computers (**hosts**) interconnected by a single technology.
- Two computers are said to be interconnected if they are able to exchange information.
- Purpose of computer networks: to **share resources** (e.g. data and applications).



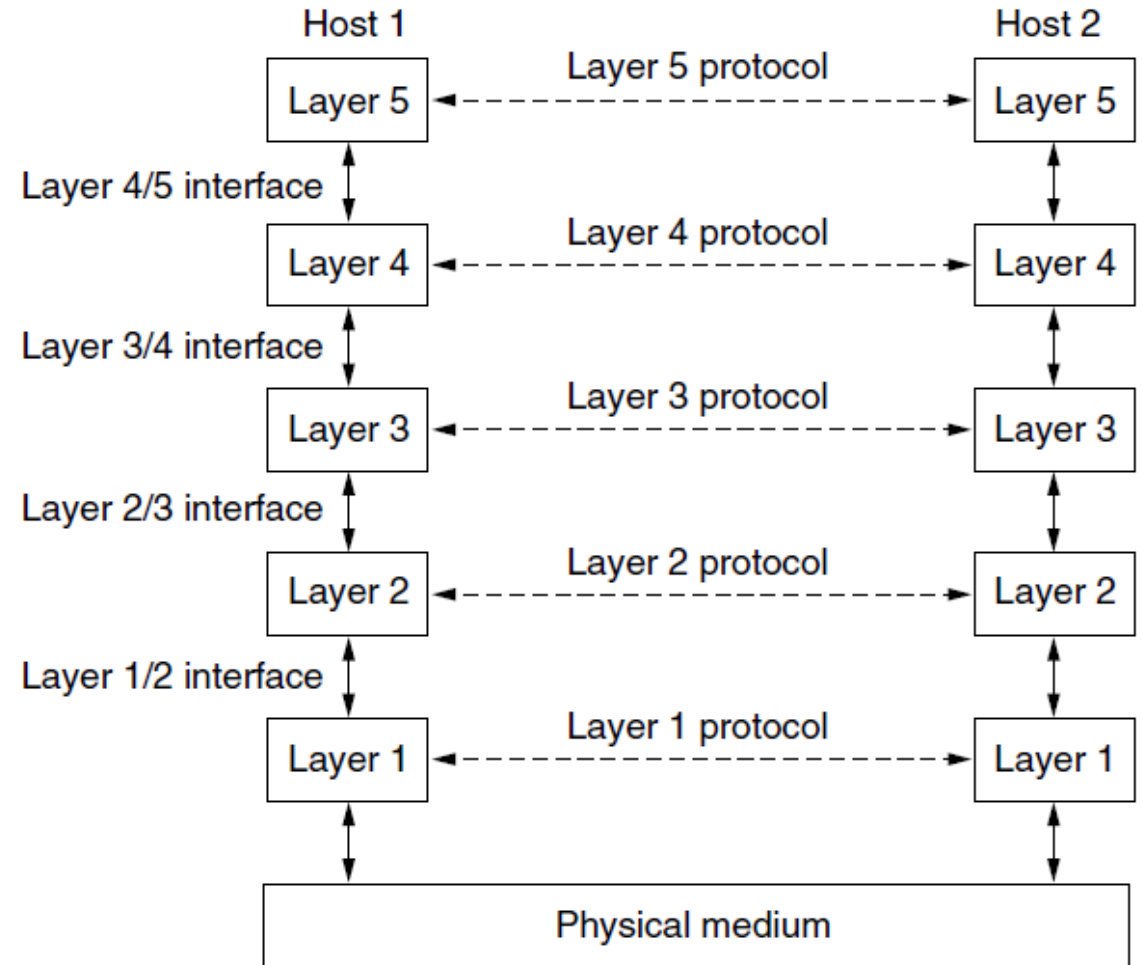
Client-server model

- **Client-server model:** a network structure that partitions tasks between the providers of a resource or service, called servers, and service requesters, called clients.
- The server is often designed to operate as a centralized system that serves many clients. The computing power, memory and storage requirements of a server must be scaled appropriately to the expected workload.



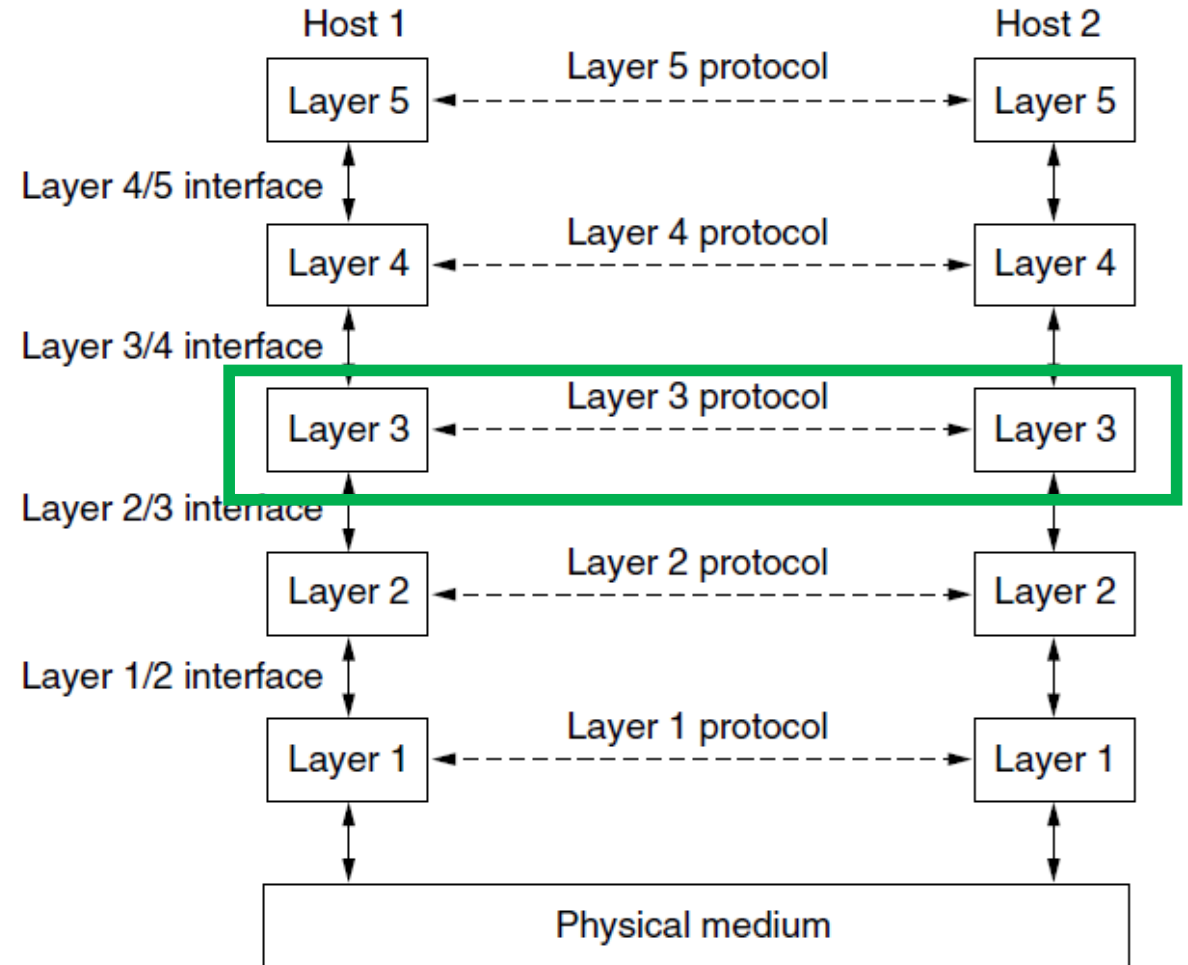
Network software: protocol hierarchies

- To reduce their design complexity, most networks are organized as a stack of **layers** or **levels**, each one built upon the one below it.
- The purpose of each layer is to offer certain services to the higher layers while shielding those layers from the details of how the offered services are actually implemented.



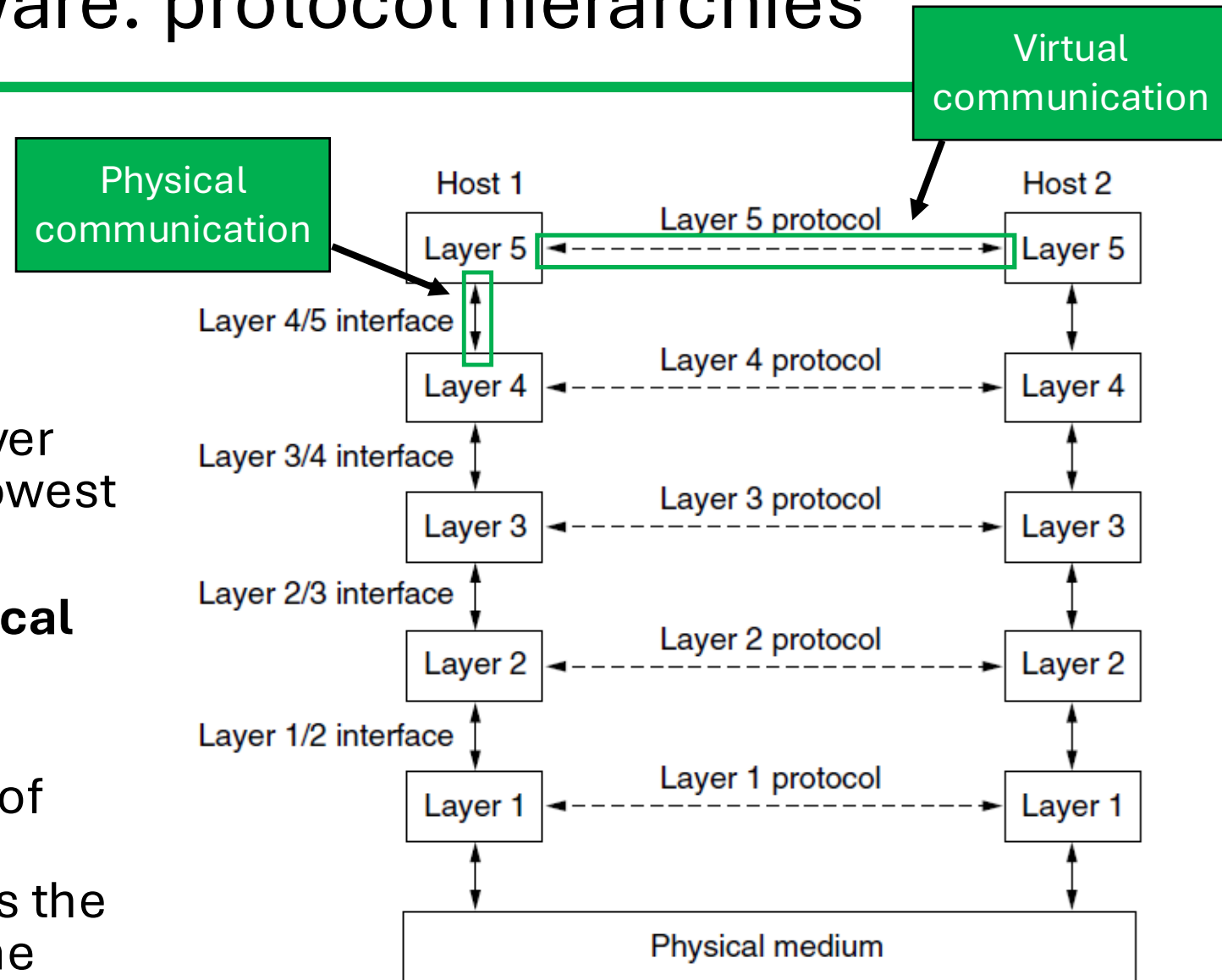
Network software: protocol hierarchies

- Two entities of the same layer on different machines are called **peers**.
- When two peers of layer n communicate, the rules and conventions used in this conversation constitute the layer n **protocol**.
- **Protocol**: agreement between the communicating parties on how communication must work.
- Layers and protocols form the **network architecture**.



Network software: protocol hierarchies

- In reality, no data are directly transferred from layer n on one machine to layer n on another machine.
- Each layer passes data to the layer immediately below it, until the lowest layer is reached.
- Layer 1 passes data to the **physical medium** through which actual communication occurs.
- An **interface** between each pair of adjacent layers defines which primitive operations and services the lower layer makes available to the upper one.



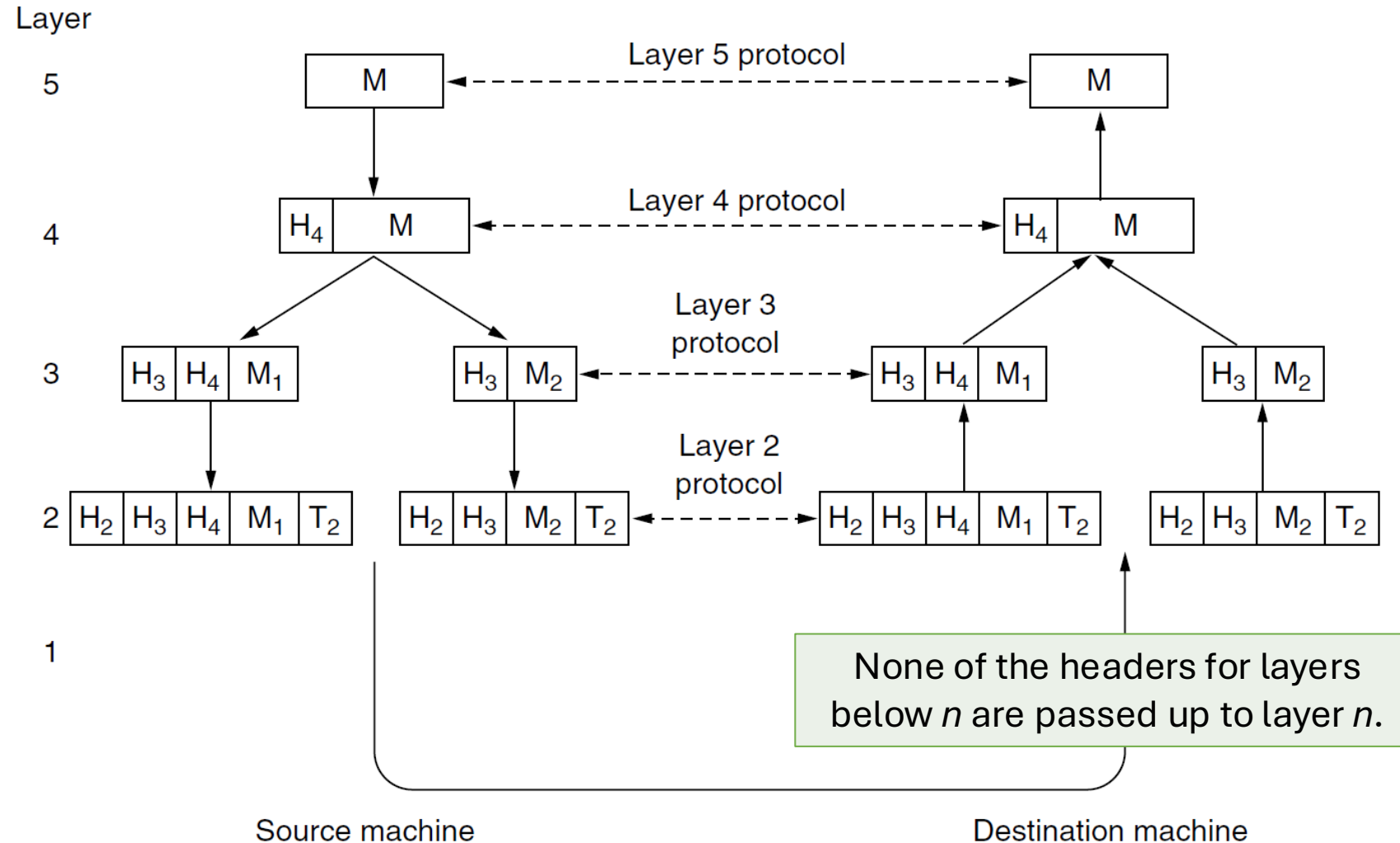
Example of network architecture

A message M is produced by an application running on layer 5 and given to layer 4 for transmission.

Layer 4 puts a header (e.g. address) in front of M to identify it and then pass it to layer 3.

Layer 3 breaks M into smaller packets (M1 and M2) and add headers to each packet which are passed to layer 2.

Layer 2 adds a header and a trailer to each packet and then pass them to layer 1 for physical transmission to the destination machine.



Services and primitives

- **Service:** a set of primitives (operations) that a layer provides to the layer above.
- What operations the layer is prepared to perform on behalf of its users
 - No information about how these operations are implemented.
 - It relates to an interface between two layers, with the lower layer being the service provider and the upper layer being the service user.

Examples of primitives for a simple connection-oriented service

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
ACCEPT	Accept an incoming connection from a peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Connection-oriented vs connectionless services

Two types of service:

- **Connection-oriented service:** the service user first establishes a connection with the receiver, then it uses the connection to send data, and finally it releases the connection (like in a telephone communication).
 - In most cases the order of data bits is preserved → bits arrive in the same order they were sent.
- **Connectionless service:** each message carries the full destination address and each one is sent independently on the subsequent messages (like in postal system).
 - Normally, when two messages are sent to the same destination, the first one sent will be the first one to arrive. However, it is possible that the first one sent can be delayed so that the second one arrives first.
 - The messages in a connectionless service are also called **datagram**.

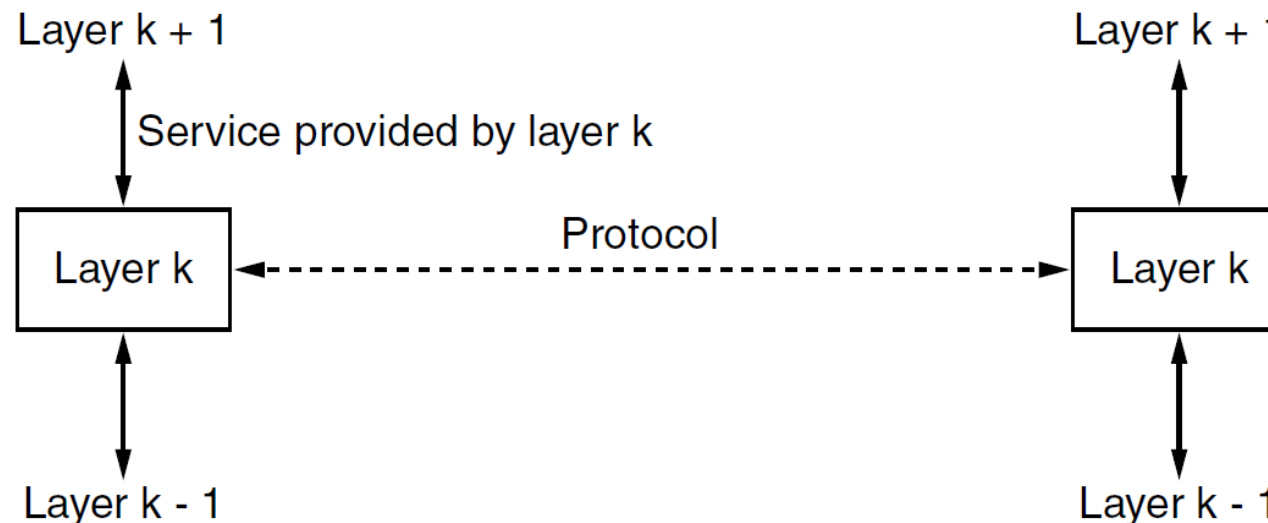
Reliable vs. unreliable services

- A **reliable** service never loses data: the receiver acknowledges the receipt of each message so the sender is sure that it arrived.
- The acknowledgement process introduces overhead and delays, sometimes undesirable.
 - E.g. in a telephone communication it is less disruptive to hear a bit of noise on the line from time to time than to experience a delay waiting for acknowledgements.
- An **unreliable** service does not return an acknowledgement to the sender

		Service	Example
Connection-oriented	{	Reliable message stream	Sequence of pages
		Reliable byte stream	Movie download
		Unreliable connection	Voice over IP
Connection-less	{	Unreliable datagram	Electronic junk mail
		Acknowledged datagram	Text messaging
		Request-reply	Database query

Protocol vs. service

- **Protocol:** The set of rules governing the format and meaning of the packets, or messages that are exchanged by the peer entities within a layer.
- **Service:** the set of primitives (operations) that a layer provides to the layer above it. They are related to the interfaces between layers.
- The protocol of layer k defines the implementation details of services which layer k provides to users of layer k+1; such details are not visible to users of layer k+1.



The TCP/IP reference model

Application layer

Transport layer

Network layer

Link layer

Physical layer

- **Worldwide Internet:** the global system of interconnected computer networks that uses the Internet protocol suite (TCP/IP) to communicate between networks and devices.
- The reference model used by the worldwide Internet is called **TCP/IP reference model** and it was introduced by Cerf and Kahn in 1974.

The TCP/IP reference model

Application layer



Transport layer

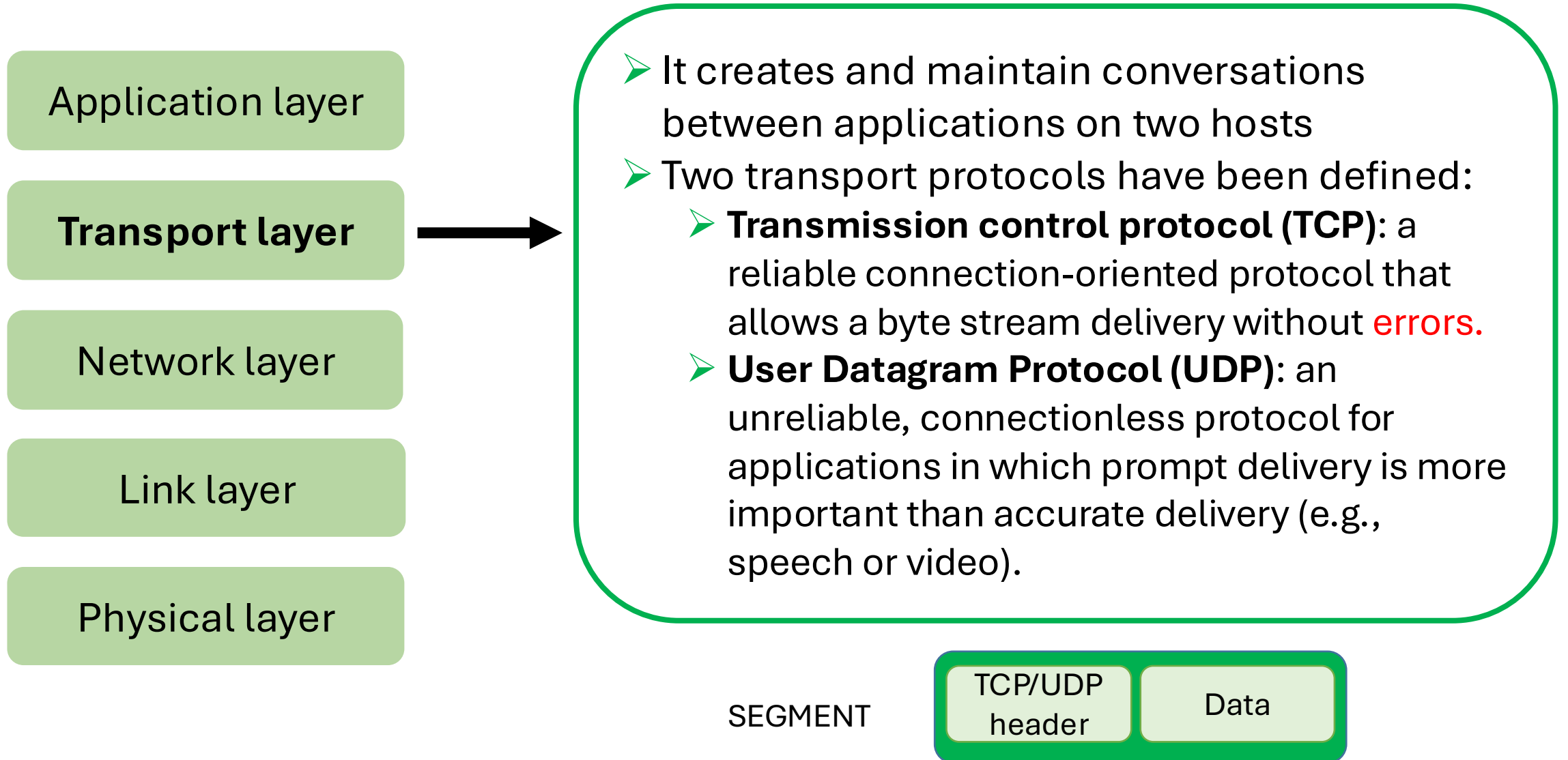
Network layer

Link layer

Physical layer

- It includes the protocols used by applications running on different hosts to communicate
- Example: A Web browser can send requests to a Web server using the **HTTP protocol**, which is a protocol of the application layer

The TCP/IP reference model



The TCP/IP reference model

Application layer

Transport layer

Network layer →

Link layer

Physical layer

- Finds the correct routing path, handles congestions and quality of service.
- Packets are sent independently to the destination by a certain path.
- This layer works with two protocols:
 - **Internet protocol (IP)**: it defines the IP addresses, i.e., that label (e.g., 192.0.2.1) that uniquely identifies a host connected to a computer network using the IP protocol.
 - **Internet Control Message Protocol (ICMP)**: it sends error and control messages.

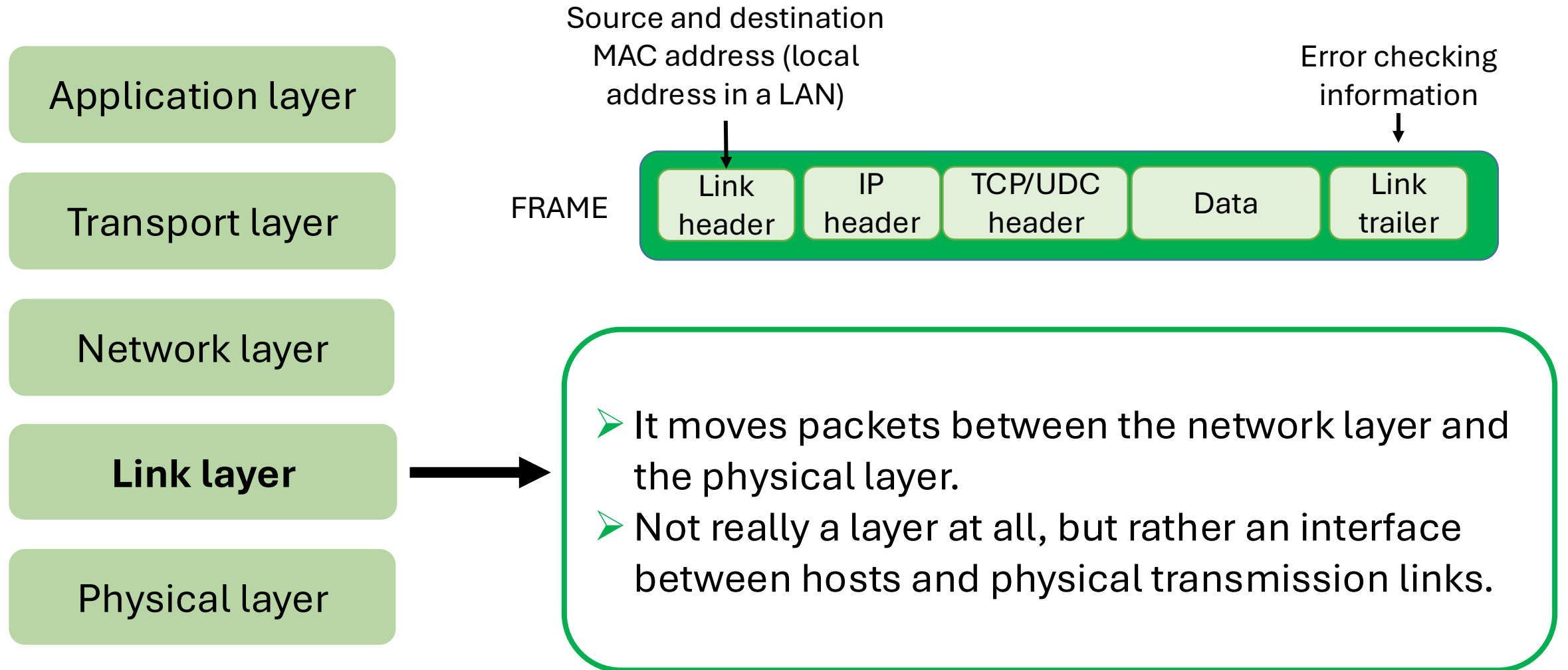
PACKET

IP
header

TCP/UDC
header

Data

The TCP/IP reference model



The TCP/IP reference model

Application layer

Transport layer

Network layer

Link layer

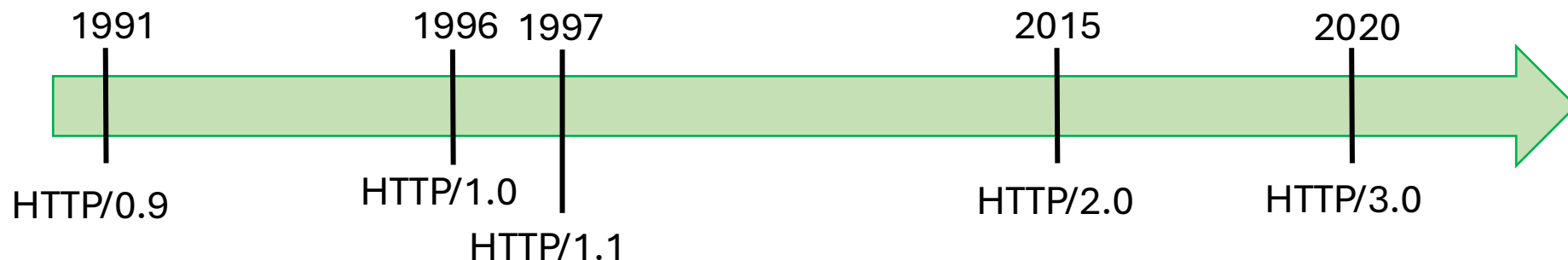
Physical layer

00010101110110101001100110101111010

- It physically receives and transmit raw bits over a communication channel.
- A standardized interface to the transmission medium (electrical, optical, or wireless communication).

The HyperText Transfer Protocol (HTTP)

- **HyperText Transfer Protocol (HTTP):** the main web communication protocol for exchanging information between servers and clients.
- Simple **request-response protocol**, proper of the application layer, running over TCP.
- It specifies what messages clients may send to servers and what responses they get back in return.
- Most of Web currently use HTTP/1.1 and HTTP/2.0, which share the same syntax.



Request structure (HTTP/1.1)

- Each **request** consists of one or more lines of ASCII text

Case-sensitive name of the method
(type of the request)

- First line:
METHOD resource protocol
Name of the resource in the server
Name of the protocol version (e.g. HTTP/1.1)

- Request headers:
Host: servername → Additional information related to the request.
In HTTP/1.1 there must be at least one header containing the DNS name of the server.

Other optional headers

- Empty line
- Request body (payload):

Optional message body

HTTP methods

- HTTP methods indicate the **desired action** to be performed for a given resource

Method	Desired action
GET	Retrieves a resource
HEAD	Retrieves a resource header
POST	Submits an entity to the specified resource (causes a change of server state)
PUT	Replaces all current representations of the target resource with the request payload.
DELETE	Deletes a resource
TRACE	Echoes the incoming request
CONNECT	Establishes a tunnel to the server identified by the target resource
OPTIONS	Describes the communication option for a resource
PATCH	Applies partial modifications to a resource

Examples of request headers

- *Host*: the server's DNS name
- *Authorization*: needed for pages that are protected. The client might need to demonstrate it has the right to see the requested page.
- *Content-Type*: the type of the content in the request body.

(Other) Examples of request headers

BONUS

- *User-Agent*: it allows the client to inform the server about the user agent making the request, e.g., the browser implementation
- *Accept*: it allows to tell the server what the client is willing to accept
- *If-Modified-Since*: time and date of last page modification, it allows the client to check the validity of cached pages (the client stores recently requested contents in a cache memory, similarly to what the server does).
- *Cache-Control*: it specifies directives that must be obeyed by all caching mechanisms along the request/response chain (e.g., *no-cache*).
- ...

Response structure (HTTP/1.1)

- A response message is sent by a server to a client as a reply to its former request message.

Three-digit integer code representing the result of the server's attempt to understand and satisfy the request.

- Status line:

Protocol status_code reason_phrase

Optional. If the status code indicated a problem, the *reason phrase* might displayed the further information about the nature of the problem.

- Response headers:

Optional headers

- Empty line

- Request body (payload):

Optional message body

Status code	Description
1xx	Informational: the request was received and is being processed,
2xx	Successful: the request was successfully received, understood, and accepted.
3xx	Redirecting: further action needs to be taken in order to complete the request.
4xx	Client error: the request contains bad syntax or cannot be fulfilled.
5xx	Server error: The server failed to fulfill an apparently valid request.

Examples of response headers

- *Date*: date and time the message was sent
- *Content-Type*: the type of the returned content (e.g., HTML, image, plain text, ...)
- *Content-Length*: the content length in bytes
- *Last-Modified*: time and date the page was last changed
- ...

(Other) Examples of response headers

BONUS

- *Content-Language*: the natural language used in the page
- *Expires*: Time and date when the page stops being valid
- *ETag*: tag for the content of the page (used for caching)
- *Location*: redirect the recipient to a location other than the requested URL for completion of the request or identification of a new resource
- ...

GET

➤ **GET:** Retrieves a resource.

GET / HTTP/1.1

Host: www.example.com

Accept-Language: en

Request example:

```
GET / HTTP/1.1
Host: www.example.com
User-Agent: Mozilla/5.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
Accept-Language: en-GB,en;q=0.5
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
```

Response example:

```
HTTP/1.1 200 OK
Date: Mon, 23 May 2005 22:38:34 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 155
Last-Modified: Wed, 08 Jan 2003 23:11:55 GMT
Server: Apache/1.3.3.7 (Unix) (Red-Hat/Linux)
ETag: "3f80f-1b6-3e1cb03b"
Accept-Ranges: bytes
Connection: close

<html>
  <head>
    <title>An Example Page</title>
  </head>
  <body>
    <p>Hello World, this is a very simple HTML document.</p>
  </body>
</html>
```

POST

- **POST:** Submits an entity to the specified resource. Can be used for creating a resource.
- If the request is accepted, then the server does something with the data that depends on the URL, and send a response page indicating the result.
 - POST is commonly used when forms are submitted, or when uploading files to servers, e.g., to extend a database appending some new data.

POST HTML Form Example

```
POST /echo/post/form HTTP/1.1
Host: reqbin.com
Content-Type: application/x-www-form-urlencoded
Content-Length: 23

key1=value1&key2=value2
```

POST JSON Data Example

```
POST /echo/post/json HTTP/1.1
Host: reqbin.com
Accept: application/json
Content-Type: application/json
Content-Length: 81

{
  "Id": 78912,
  "Customer": "Jason Sweet",
  "Quantity": 1,
  "Price": 18.00
}
```

Other important methods

- **PUT:** Replaces all current representations of the target resource with the request payload. Can be used for updating/modifying a resource.
- **DELETE:** Deletes a resource.
- **PATCH:** Applies partial modifications to a resource.
- **HEAD:** it just asks for the message header, without the actual content. It can be used to collect information for indexing purposes, or to test a resource for validity.

POST vs PUT vs PATCH

➤ Difference between POST and PUT?

- **POST**: the client asks to the server to accept the entity enclosed in the request as a new subordinate of the resource identified by the request URL.
- **PUT**: the client asks to the server to store the enclosed entity under the request URL. → If the request URL refers to an already existing resource, the server replaces the existing entity with the new enclosed entity. If the URL does not point to an existing resource, the server can create the resource with that URL.

➤ Difference between PUT and PATCH?

- **PUT** supplies a modified version of the resource to be allocated at the request URL.
- **PATCH** supplies a set of instructions to modify the resource at the requested URL.

Other (less) important methods

BONUS

- **TRACE:** it asks the server to send back the request. Useful when requests are not processed correctly to check what the server received.
- **CONNECT:** it lets a user make a connection to a Web server through an intermediate device, such as a Web cache.
- **OPTIONS:** it queries the server for a resource and obtain the methods and headers that can be used with that resource.

Properties of the HTTP methods

- **Safe methods:** A request method is safe if a request with that method has no intended effect on the server.
 - GET, HEAD, OPTIONS, and TRACE are safe, since they only read data from the server.
 - POST, PUT, DELETE, PATCH, CONNECT are not safe, since they may modify the server state.
- **Idempotent methods:** A request method is idempotent if multiple identical requests with that method have the same intended effect as a single such request.
 - PUT and DELETE, and safe methods are idempotent.
 - POST, PATCH and CONNECT are not necessarily idempotent. Sending an identical POST request multiple times may further modify the state of the server.
- **Cacheable methods:** A request method is cacheable if responses to requests with that method may be stored by the client in the cache for future reuse.
 - Only GET, HEAD, and POST are cacheable.

Summary of properties of the HTTP methods

METHOD	REQUEST HAS BODY	RESPONSE HAS BODY	IS SAFE	IS IDEMPOTENT	IS CACHEABLE
GET	Optional	Yes	Yes	Yes	Yes
HEAD	Optional	No	Yes	Yes	Yes
POST	Yes	Yes	No	No	Yes
PUT	Yes	Yes	No	Yes	No
DELETE	Optional	Yes	No	Yes	No
TRACE	No	Yes	Yes	Yes	No
CONNECT	Optional	Yes	No	No	No
OPTIONS	Optional	Yes	Yes	Yes	No
PATCH	Yes	Yes	No	No	No

Example: inspecting the HTTP requests in Chrome

The screenshot shows a web browser with the address bar at `gcappon.github.io/teaching/`. The page content includes a profile for Giacomo Cappon, an Assistant Professor at the University of Padova, and two course listings for 'Biomedical Wearable Technologies for Healthcare and Wellbeing' for the years 2023/24 and 2022/23. The 'Network' tab in the Chrome DevTools is open, showing a list of requests. The 'teaching/' request is selected, and its details are visible on the right. The 'Request Headers' tab is active, showing the request method as GET and the status code as 200 OK. The 'Response Headers' tab is also visible, showing various headers like 'Accept-Ranges', 'Access-Control-Allow-Origin', 'Age', 'Cache-Control', 'Content-Encoding', 'Content-Length', 'Content-Type', 'Date', 'Etag', and 'Expires'.

Request headers

Requested page and contents

Response headers

Teaching

[Biomedical Wearable Technologies for Healthcare and Wellbeing, 2023/24](#)

Masters course, *University of Padova, Department of Information Engineering*, 2024

Professor for the course unit Biomedical Wearable Technologies for Healthcare and Wellbeing of the Bioengineering course (Master's degree), School of Engineering, University of Padova (6 credits/CFU, SSD ING/INF-06) (~120 students).

[Biomedical Wearable Technologies for Healthcare and Wellbeing, 2022/23](#)

Masters course, *University of Padova, Department of Information Engineering*

Contract professor for the course unit Biomedical Wearable Technologies for Healthcare and Wellbeing of the Bioengineering course (Master's degree), School of Engineering, University of Padova (3 credits/CFU, SSD ING/INF-06)

Giacomo Cappon

Assistant Professor @ DEI, University of Padova, YOLO committer

Padova, Italy

Email

ResearchGate

Github

Google Scholar

PubMed

ORCID

Network

teaching/

Request URL: `https://gcappon.github.io/teaching/`

Request Method: GET

Status Code: 200 OK

Remote Address: 185.199.110.153:443

Referrer Policy: strict-origin-when-cross-origin

Response Headers

Accept-Ranges: bytes

Access-Control-Allow-Origin: *

Age: 0

Cache-Control: max-age=600

Content-Encoding: gzip

Content-Length: 2891

Content-Type: text/html; charset=utf-8

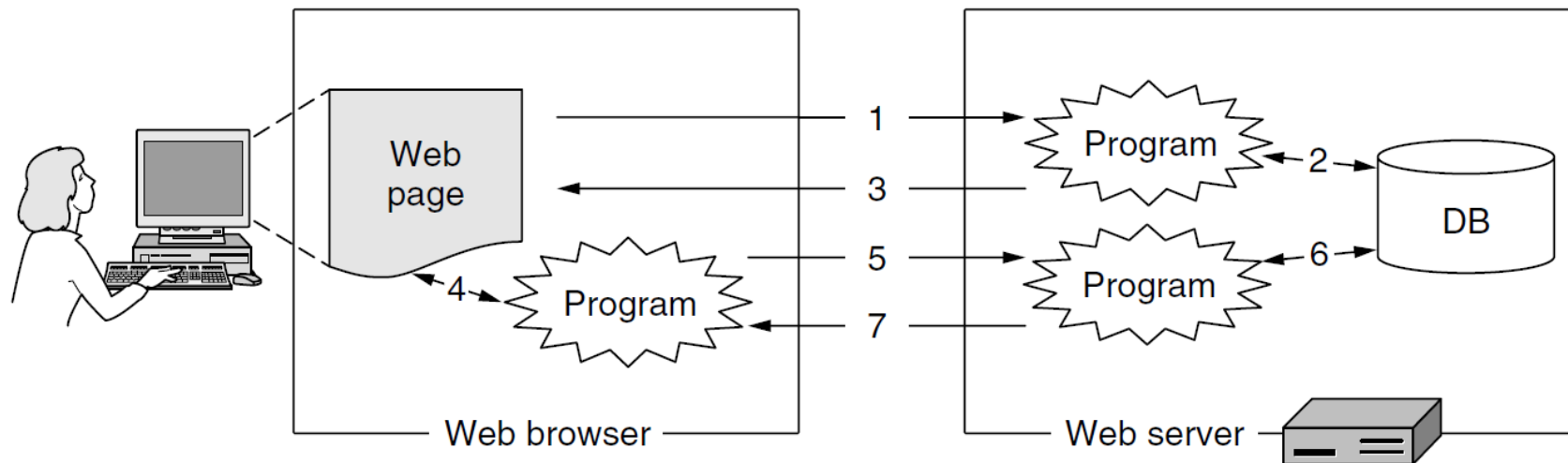
Date: Thu, 01 Feb 2024 12:49:10 GMT

Etag: W/"64d79e9b-293c"

Expires: Thu, 01 Feb 2024 12:59:10 GMT

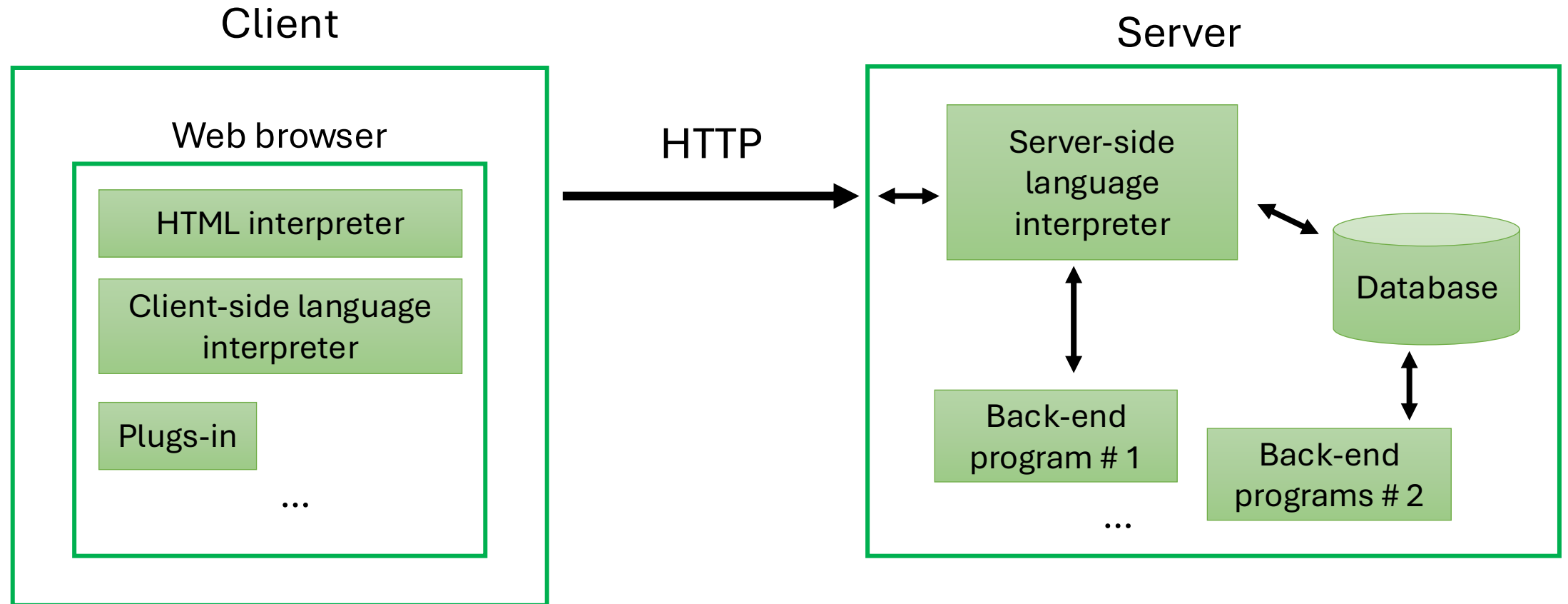
Web application

- **Web application:** a Web page or a collection of Web pages, delivered over HTTP, which use server-side or client-side processing, i.e., programs, to provide an "application-like" experience within a Web browser.
- Web applications access data that are stored in databases in the server and use Web protocols (e.g., HTTP) to exchange data with the client. They use the Web browser to display a user interface by which the user can use the app (making queries, inserting data, view retrieved data, etc.).
- Advantage of Web applications: The users do not need to install separate application programs, and data can be accessed from different computers and backed up by the service operator.
- Examples of Web apps: Gmail, Google Docs, Trello, online banking apps, online retail sales apps, ...



To act as applications, Web pages can no longer be static, but they need to generate **dynamic content**.

Example of a Web application architecture



Structured data formats

- Web applications need to exchange data with the server.
- HTML is not suitable for representing data because it mixes content with formatting. It is indeed concerned about the presentation of the content.
 - It is difficult to automatically retrieve data within an HTML file
- Two popular structured data formats that facilitate the automatic structured data processing:
 - Extensible Markup Language (XML)
 - JavaScript Object Notation (JSON)

Extensible Markup Language (XML)

- A language for specifying structured content.
- Introduced in 1998.
- It defines a set of rules for encoding documents in a format that is both human-readable and machine-readable.
- Unlike HTML, there are no defined tags for XML. Each user can define their own tags.
- Extension .xml

Example of XML code that define a book list

```
<?xml version="1.0" ?>
<book list>
  <book>
    <title> Human Behavior and the Principle of Least Effort </title>
    <author> George Zipf </author>
    <year> 1949 </year>
  </book>
  <book>
    <title> The Mathematical Theory of Communication </title>
    <author> Claude E. Shannon </author>
    <author> Warren Weaver </author>
    <year> 1949 </year>
  </book>
  <book>
    <title> Nineteen Eighty-Four </title>
    <author> George Orwell </author>
    <year> 1949 </year>
  </book>
</book list>
```

JavaScript Object Notation (JSON)

- A format that uses human-readable text to store and transmit data objects consisting of attribute–value pairs and arrays.
- Introduced in 2001.
- JSON is language-independent. It was derived from JavaScript, but many modern programming languages include code to generate and parse JSON-format data.
- Extension .json
- JSON data types: number, string, Boolean, null (empty value)
- JSON data structures
 - Object
 - Array

JSON data structures

- **JSON object:** a collection of name–value pairs where the names (also called keys) are strings. The order of pairs is not relevant. The object is surrounded by curly braces.

```
{
  "employee": {
    "id": 1,
    "name": "Admin",
    "location": "USA"
  }
}
```

Objects can contain array and arrays can contain objects.

- **JSON array:** an ordered list of zero or more elements, each of which may be of any type. Arrays use square bracket notation with comma-separated elements.

```
{
  "employees": [
    {
      "id": 1,
      "name": "Admin",
      "location": "India"
    },
    {
      "id": 2,
      "name": "Author",
      "location": "USA"
    },
    {
      "id": 3,
      "name": "Visitor",
      "location": "USA"
    }
  ]
}
```

Why JSON are important here? What we need?

- The Fitbit data you will need to develop your project (stored in the IMPACT server are stored in a JSON format
- We need to learn how to make and "debug" our requests!



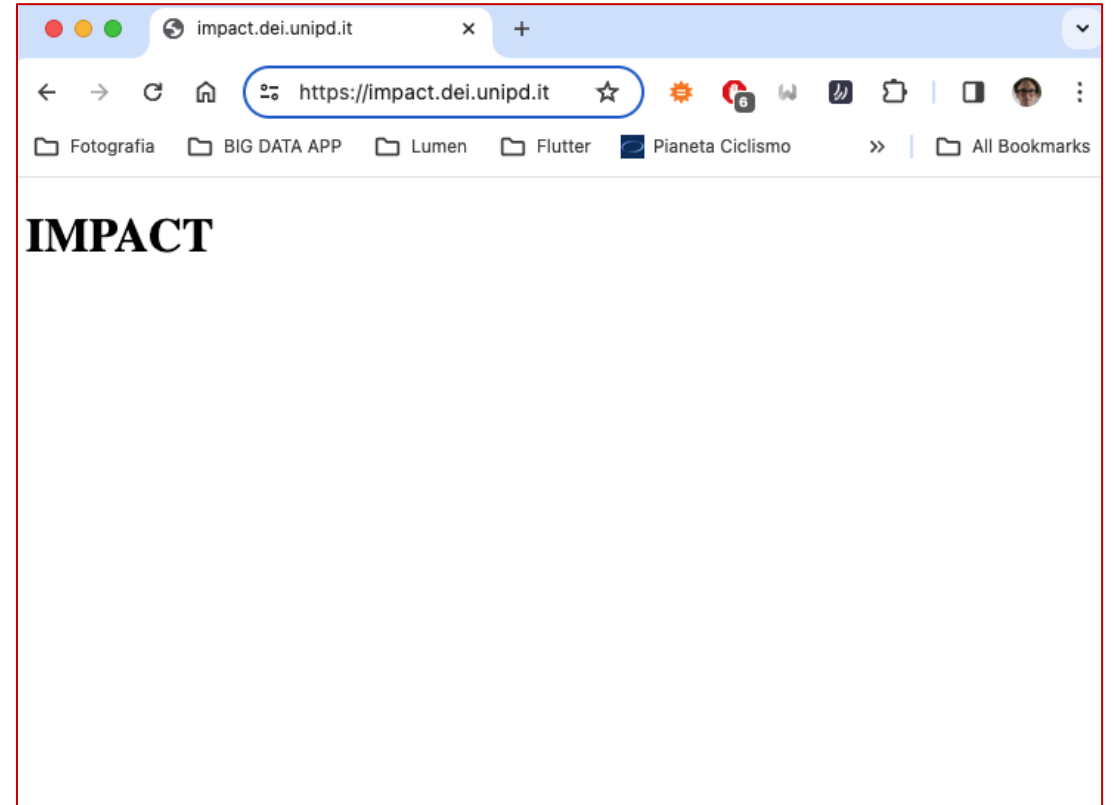
POSTMAN

- We saw that using a browser we are actually visualizing the response of an HTTP request using the GET method.
- What if we need to make an authenticated request?
What if we need to make a POST request?
- POSTMAN can help us here:
<https://www.postman.com>
- We will use it during the theory lessons to try what we learn



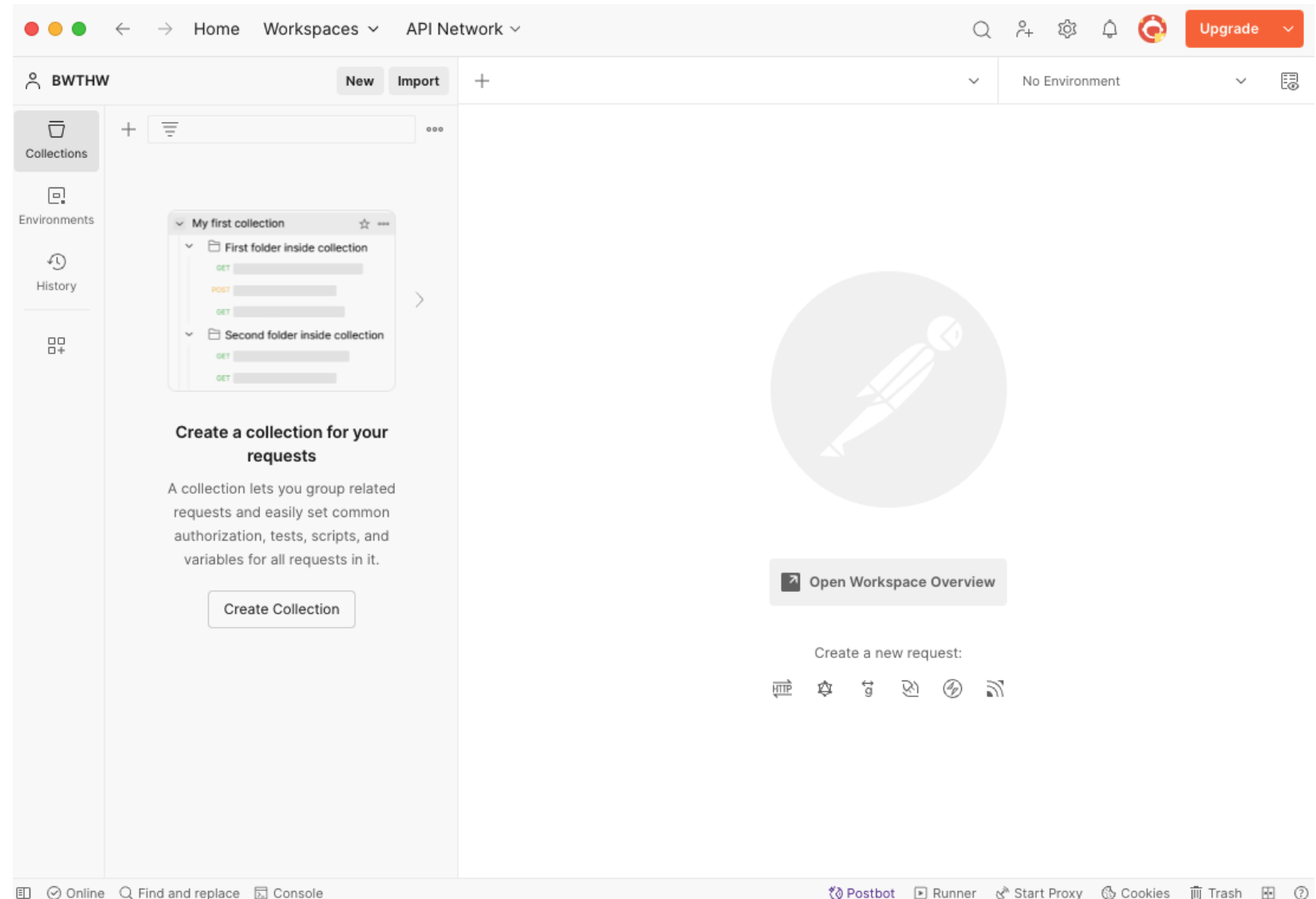
Hello IMPACT

- At <https://impact.dei.unipd.it/> you will obtain the following
- Let's replicate it in POSTMAN



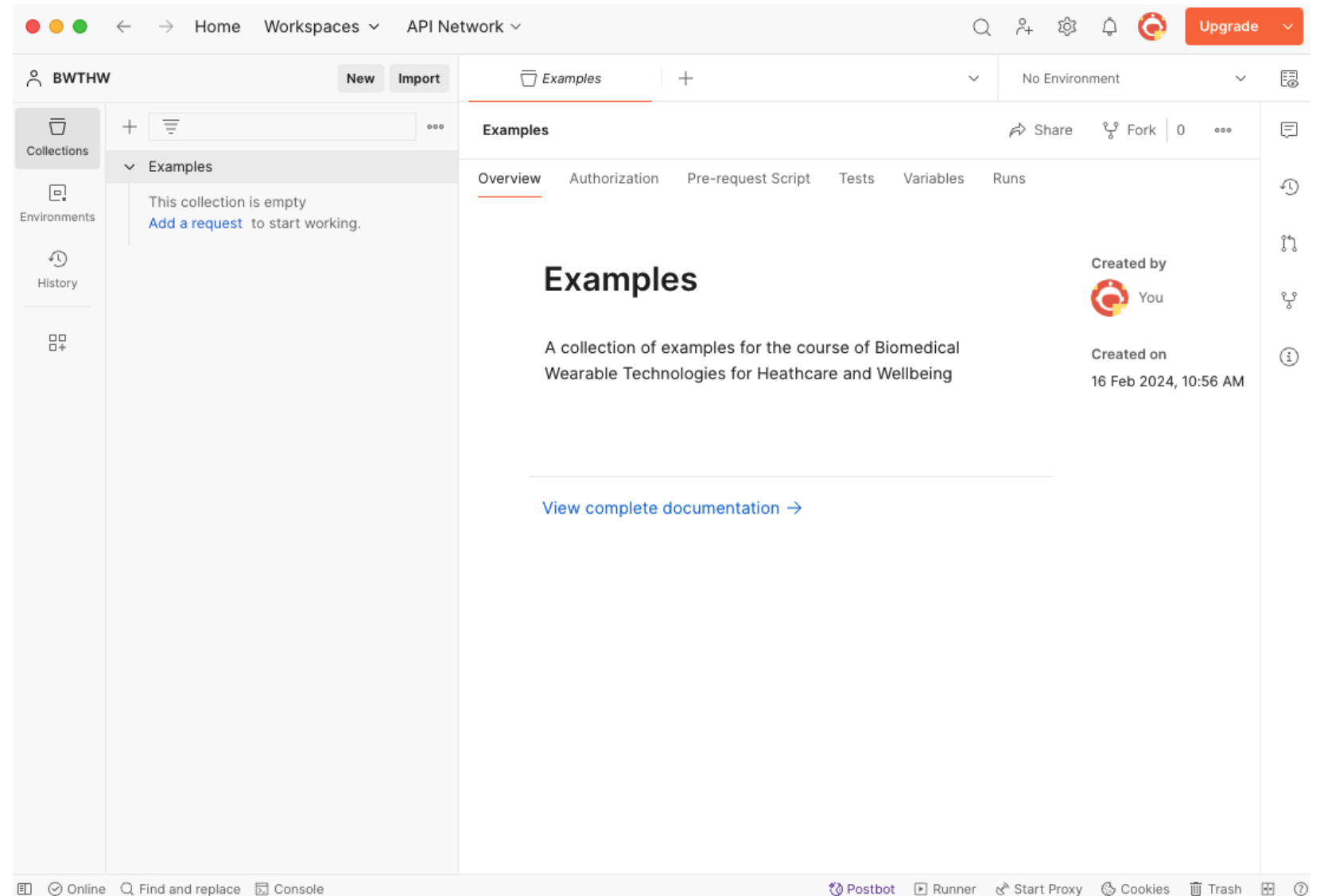
Hello IMPACT

- This is the POSTMAN interface
- Many things can be done (e.g., automatic debugging, writing custom scripts, etc...)
- We will limit to create and organize a collection of HTTP requests



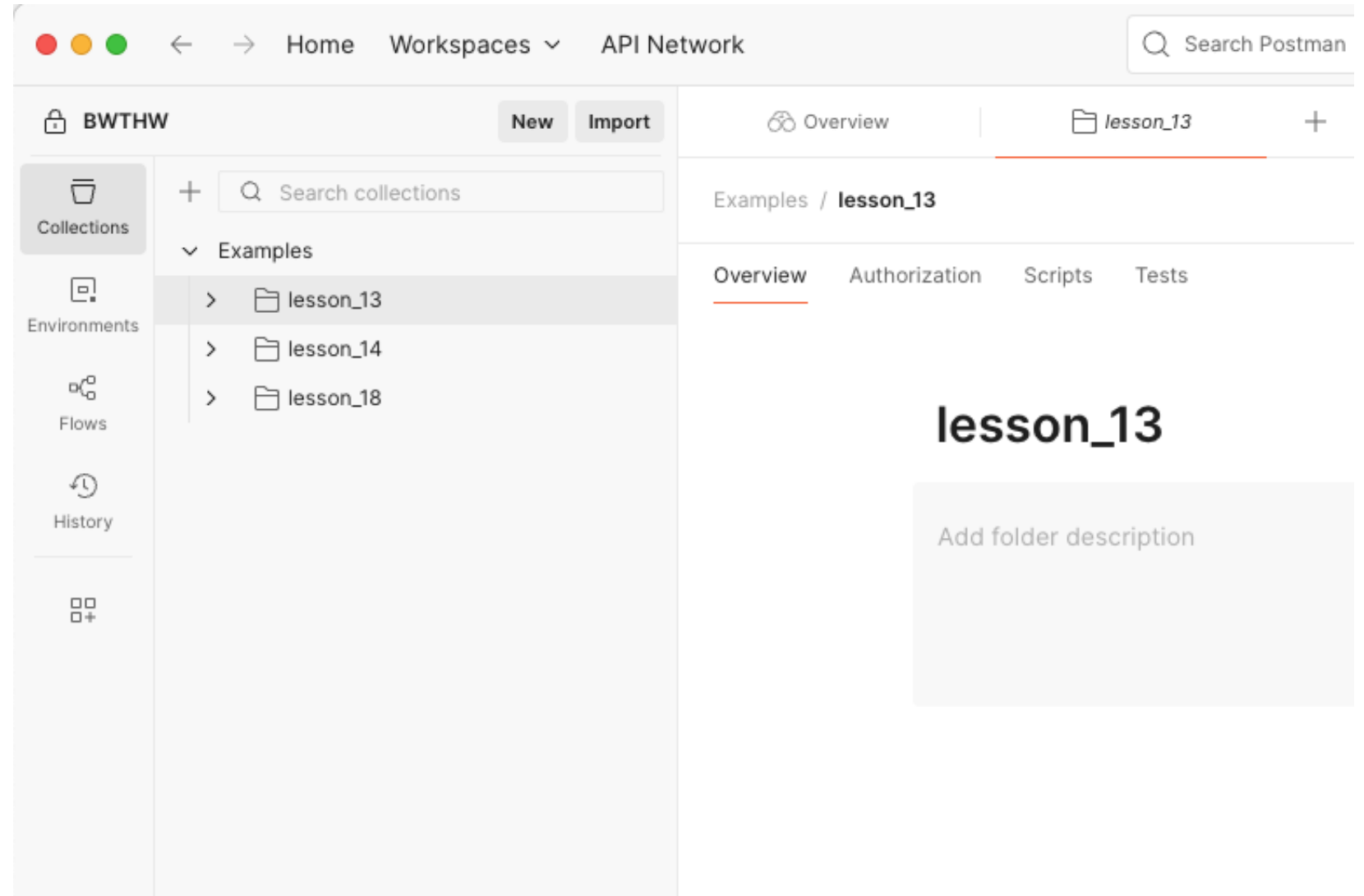
Hello IMPACT

- Go to the "Collections" tab
- Press the "+" button
- Create a "Blank collection"
- Call it "Examples": here we will collect all the theory lessons examples



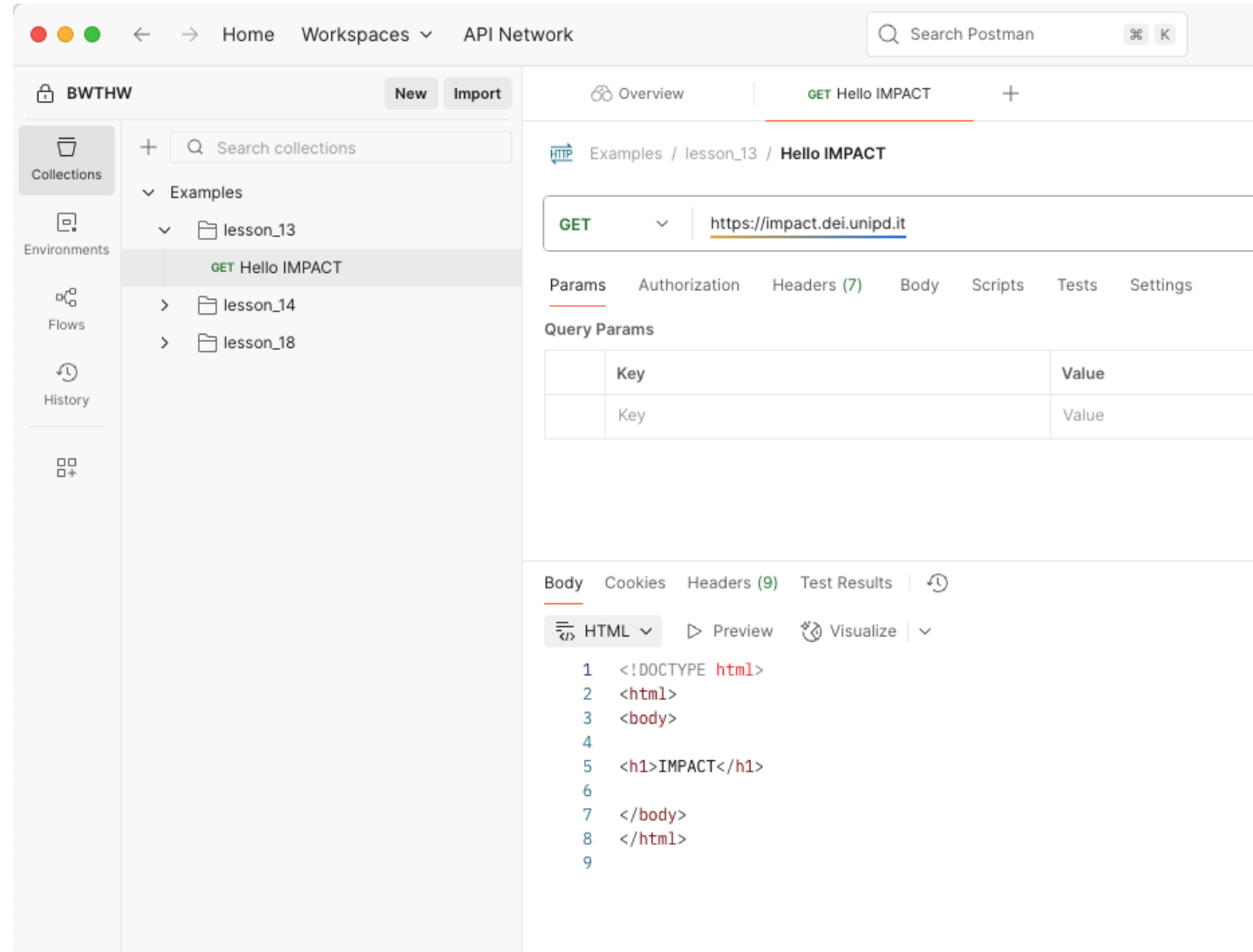
Hello IMPACT

- Right click the “Examples” collection
- Select “Add folder”
- Call it as you want, e.g.: “lesson_13”



Hello IMPACT

- Right click the folder and choose “Add request”
- Call it as you want, e.g., “Hello IMPACT”
- Choose the GET method
- And write the IMPACT home URL, i.e., <https://impact.dei.unipd.it/>
- Click “Send” and inspect the response



References

- Tanenbaum, Wetherall – Computer Networks – Fifth Edition
 - Chapter 1 – Introduction
 - Chapter 7 – The application layer
- POSTMAN: <https://www.postman.com>